

AN EXACT DIVISOR PARAMETRISATION FOR THE INTEGER SOLUTIONS OF $z^2 + y^2z + x^3 - 2 = 0$

ABSTRACT. We give a complete elementary solution of the Diophantine equation

$$z^2 + y^2z + x^3 - 2 = 0, \quad x, y, z \in \mathbb{Z}.$$

The key point is the factorisation

$$z^2 + y^2z = z(z + y^2).$$

Since the case $z = 0$ is impossible, every solution is obtained by choosing an integer x , a non-zero divisor z of $2 - x^3$, and then requiring

$$\frac{2 - x^3}{z} - z$$

to be a non-negative square. This gives an exact finite divisor test for each fixed value of x , and no external results are required.

1. STATEMENT OF THE RESULT

We solve the equation

$$(1) \quad z^2 + y^2z + x^3 - 2 = 0, \quad x, y, z \in \mathbb{Z}.$$

The solution is most naturally expressed as a divisor parametrisation. Throughout the paper, a *non-negative square* means an integer of the form r^2 with $r \in \mathbb{Z}$.

Theorem 1 (Complete divisor parametrisation). *The integer solutions of (1) are exactly the triples*

$$(2) \quad (x, y, z) \in \mathbb{Z}^3$$

with the following three properties:

- (i) $z \neq 0$;
- (ii) $z \mid (2 - x^3)$;
- (iii) *the integer*

$$Q(x, z) := \frac{2 - x^3}{z} - z$$

is a non-negative square, and y is one of its square roots.

Equivalently,

$$(3) \quad \boxed{\mathcal{S} = \left\{ (x, y, z) \in \mathbb{Z}^3 : z \neq 0, \ z \mid (2 - x^3), \ y^2 = \frac{2 - x^3}{z} - z \right\}.}$$

In particular, for fixed x one obtains the complete finite list of solutions by checking the finitely many divisors z of $2 - x^3$.

The following equivalent form is often cleaner conceptually.

Theorem 2 (Factor-pair form). *The integer solutions of (1) are in bijection with the quadruples*

$$(x, a, b, y) \in \mathbb{Z}^4$$

satisfying

$$(4) \quad ab = 2 - x^3, \quad b - a = y^2, \quad a \neq 0.$$

The bijection sends a solution (x, y, z) to

$$(x, a, b, y) = (x, z, z + y^2, y),$$

and the inverse map sends (x, a, b, y) to

$$(x, y, z) = (x, y, a).$$

Equivalently, one may choose a factor pair (a, b) of $2 - x^3$ whose difference $b - a$ is a non-negative square, and then set

$$z = a, \quad y = \pm\sqrt{b - a},$$

with the convention that if $b - a = 0$, then the single value $y = 0$ is obtained.

2. PROOF OF THE PARAMETRISATION

We begin with the only exceptional case that could obstruct division by z .

Lemma 3. *If $(x, y, z) \in \mathbb{Z}^3$ satisfies (1), then $z \neq 0$.*

Proof. Suppose, for contradiction, that $z = 0$. Substituting into (1) gives

$$x^3 - 2 = 0,$$

so $x^3 = 2$. But no integer cube is equal to 2: indeed, if $x \leq 1$, then $x^3 \leq 1$, while if $x \geq 2$, then $x^3 \geq 8$. This contradiction proves $z \neq 0$. $\square$

Lemma 4 (The fundamental factorisation). *For integers x, y, z , equation (1) is equivalent to*

$$(5) \quad z(z + y^2) = 2 - x^3.$$

Proof. Starting from (1), move $x^3 - 2$ to the other side:

$$z^2 + y^2z = 2 - x^3.$$

The left-hand side factors as

$$z^2 + y^2z = z(z + y^2).$$

This gives (5). Conversely, expanding (5) gives

$$z^2 + y^2z = 2 - x^3,$$

which is exactly (1). Hence the two equations are equivalent. $\square$

Lemma 5 (Divisor-square criterion). *Let $x, z \in \mathbb{Z}$ with $z \neq 0$. There exists an integer y satisfying (1) if and only if*

$$z \mid (2 - x^3)$$

and

$$\frac{2 - x^3}{z} - z$$

is a non-negative square. In that case the possible values of y are exactly the integer square roots of this number.

Proof. Assume first that there is an integer y such that (x, y, z) satisfies (1). By Lemma 4,

$$z(z + y^2) = 2 - x^3.$$

Because $z \neq 0$, this implies immediately that z divides $2 - x^3$. Dividing by z gives

$$z + y^2 = \frac{2 - x^3}{z}.$$

Therefore

$$y^2 = \frac{2 - x^3}{z} - z.$$

The right-hand side is thus a non-negative square, namely the square of the integer y .

Conversely, suppose that $z \mid (2 - x^3)$ and that

$$\frac{2 - x^3}{z} - z = r^2$$

for some integer r . Put $y = r$. Then

$$z + y^2 = z + r^2 = z + \left(\frac{2 - x^3}{z} - z \right) = \frac{2 - x^3}{z}.$$

Multiplying both sides by z yields

$$z(z + y^2) = 2 - x^3.$$

By Lemma 4, this is equivalent to (1). Thus (x, r, z) is a solution. Since the displayed equation forces $y^2 = r^2$, the only possible values of y are precisely the integer square roots of r^2 . $\square$

Proof of Theorem 1. Let (x, y, z) be an integer solution of (1). By Lemma 3, $z \neq 0$. Lemma 5 then gives

$$z \mid (2 - x^3) \quad \text{and} \quad y^2 = \frac{2 - x^3}{z} - z.$$

Thus every solution belongs to the set displayed in (3).

Conversely, suppose (x, y, z) belongs to the set displayed in (3). Then $z \neq 0$, $z \mid (2 - x^3)$, and

$$y^2 = \frac{2 - x^3}{z} - z.$$

Rearranging gives

$$z + y^2 = \frac{2 - x^3}{z}.$$

After multiplication by z ,

$$z(z + y^2) = 2 - x^3.$$

By Lemma 4, this is equivalent to (1). Hence (x, y, z) is a solution. This proves both inclusions and therefore proves the complete parametrisation.

Finally, if x is fixed, then $2 - x^3 \neq 0$, because $x^3 = 2$ has no integer solution. A non-zero integer has only finitely many divisors. Hence the stated procedure checks only finitely many possible values of z and gives the complete finite list for that fixed x . $\square$

Proof of Theorem 2. Let (x, y, z) be a solution of (1). Define

$$a = z, \quad b = z + y^2.$$

By Lemma 3, $a = z \neq 0$. By Lemma 4,

$$ab = z(z + y^2) = 2 - x^3.$$

Moreover,

$$b - a = (z + y^2) - z = y^2.$$

Therefore (x, a, b, y) satisfies (4).

Conversely, suppose $(x, a, b, y) \in \mathbb{Z}^4$ satisfies (4). Put $z = a$. Since $b - a = y^2$, we have

$$b = z + y^2.$$

The identity $ab = 2 - x^3$ becomes

$$z(z + y^2) = 2 - x^3.$$

By Lemma 4, (x, y, z) satisfies (1).

It remains only to check that the two constructions are inverse to each other. Starting with a solution (x, y, z) , the first construction gives

$$(x, a, b, y) = (x, z, z + y^2, y),$$

and the inverse sends this back to (x, y, z) . Starting with a quadruple (x, a, b, y) satisfying (4), the inverse gives (x, y, a) ; applying the first construction gives

$$(x, a, a + y^2, y).$$

But $b - a = y^2$, so $a + y^2 = b$. Hence the original quadruple is recovered. Thus the correspondence is a bijection. $\square$

3. CONSEQUENCES AND EXAMPLES

The equation is even in y , and the parametrisation makes this symmetry explicit.

Corollary 6 (Sign symmetry in y). *If (x, y, z) is an integer solution of (1), then $(x, -y, z)$ is also an integer solution. If $y \neq 0$, these are distinct solutions; if $y = 0$, they coincide.*

Proof. The equation contains y only through y^2 . Therefore replacing y by $-y$ does not change the left-hand side of (1). The final assertion is immediate. $\square$

Corollary 7 (A sign restriction for $x \geq 2$). *If $x \geq 2$ and (x, y, z) is a solution of (1), then*

$$z < 0 < z + y^2.$$

In particular, $y \neq 0$ for every solution with $x \geq 2$.

Proof. For $x \geq 2$, we have

$$2 - x^3 < 0.$$

By Lemma 4,

$$z(z + y^2) = 2 - x^3 < 0.$$

Thus the two factors z and $z + y^2$ have opposite signs. Since $y^2 \geq 0$, we have

$$z \leq z + y^2.$$

The only way two ordered real numbers $z \leq z + y^2$ can have opposite signs is

$$z < 0 < z + y^2.$$

This also implies $y^2 > 0$, so $y \neq 0$. $\square$

For illustration, the following table was obtained directly from Theorem 1. For each displayed value of x , the listed values of z are exactly those shown in the table; each row gives $Q(x, z) = y^2$ and hence the corresponding values of y .

x	$2 - x^3$	admissible z	corresponding y
1	1	1, -1	0
0	2	1, -2	± 1
-2	10	1, -10	± 3
-7	345	5, -69	± 8
8	-510	-15, -34	± 7
20	-7998	-31, -258	± 17
26	-17574	-58, -303	± 19
26	-17574	-87, -202	± 17

For example, take $x = 8$. Then

$$2 - x^3 = 2 - 512 = -510.$$

The divisor $z = -15$ gives

$$Q(8, -15) = \frac{-510}{-15} - (-15) = 34 + 15 = 49 = 7^2,$$

so

$$(8, 7, -15) \quad \text{and} \quad (8, -7, -15)$$

are solutions. Similarly, the divisor $z = -34$ gives

$$Q(8, -34) = \frac{-510}{-34} - (-34) = 15 + 34 = 49 = 7^2,$$

so

$$(8, 7, -34) \quad \text{and} \quad (8, -7, -34)$$

are also solutions.

4. VERIFICATION CERTIFICATE FOR THE SAMPLE TABLE

The table can be checked without any search. Each row follows from the identity

$$z^2 + y^2z + x^3 - 2 = 0.$$

For instance, the row $(x, y, z) = (26, 19, -58)$ gives

$$z^2 + y^2z + x^3 - 2 = (-58)^2 + 19^2(-58) + 26^3 - 2.$$

The terms are

$$(-58)^2 = 3364, \quad 19^2(-58) = 361(-58) = -20938, \quad 26^3 - 2 = 17574.$$

Therefore

$$3364 - 20938 + 17574 = 0.$$

The other listed rows are verified in the same way, or more systematically by checking the divisor-square identity

$$y^2 = \frac{2 - x^3}{z} - z.$$

This identity is equivalent to the original equation by Lemma 4, so the table is not a numerical guess but a direct certificate.

5. CONCLUSION

The factorisation

$$z^2 + y^2z = z(z + y^2)$$

reduces the original Diophantine equation exactly to a divisor-square condition. There is no missing exceptional case: $z = 0$ would force $x^3 = 2$, impossible over the integers. Hence every integer solution is obtained by choosing $x \in \mathbb{Z}$, a non-zero divisor z of $2 - x^3$, and a square root of

$$\frac{2 - x^3}{z} - z,$$

provided this number is a non-negative square. This proves the complete classification of the integer solutions of (1).